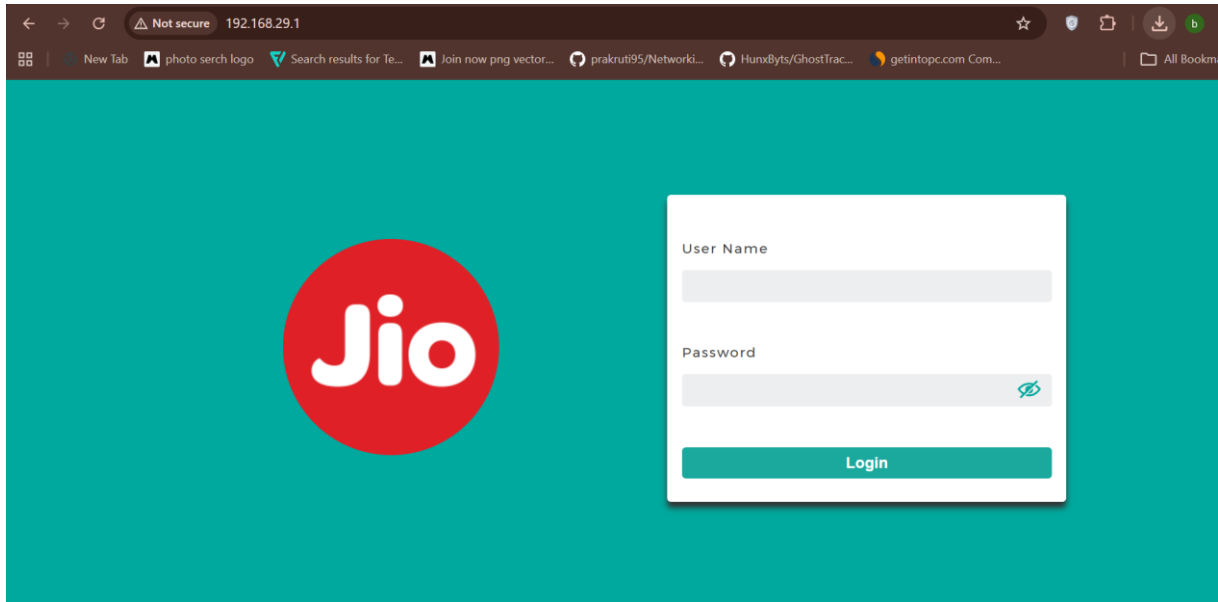


Assignment:-3

Task 1:- Log in to your home Wi-Fi router's admin panel using its default IP address (usually 192.168.0.1 or 192.168.1.1) and take a screenshot of the login page. If you don't know your router's IP, open Command Prompt and run 'ipconfig' — look for 'Default Gateway'.



Task 2:- Find and note down the current IP address, subnet mask, and default gateway assigned to your laptop or mobile device when connected to your Wi-Fi network.

Parameter	Value
IPv4 Address	192.168.29.49
Subnet Mask	255.255.255.0
Default Gateway	192.168.29.1
Connection Type	Wireless LAN (Wi-Fi)

Task 3:- Change your device's IP address from 'Obtain IP automatically' (DHCP) to a static IP in the same range as your router (for example, set it to 192.168.1.50 if your router is 192.168.1.1), and test if you still have internet access. Remember to revert to automatic/DHCP after testing to avoid network issues.

```
C:\Users\pc>ping 192.168.29.1
```

```
Pinging 192.168.29.1 with 32 bytes of data:
```

```
Reply from 192.168.29.1: bytes=32 time=1ms TTL=64
```

```
Reply from 192.168.29.1: bytes=32 time=1ms TTL=64
```

```
Reply from 192.168.29.1: bytes=32 time=1ms TTL=64
```

```
Reply from 192.168.29.1: bytes=32 time=1ms TTL=64
```

```
Ping statistics for 192.168.29.1:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
```

```
Approximate round trip times in milli-seconds:
```

```
    Minimum = 1ms, Maximum = 1ms, Average = 1ms
```

```
C:\Users\pc>ping google.com
```

```
Pinging google.com [2404:6800:4000:1025::64] with 32 bytes of data:
```

```
Reply from 2404:6800:4000:1025::64: time=20ms
```

```
Reply from 2404:6800:4000:1025::64: time=20ms
```

```
Reply from 2404:6800:4000:1025::64: time=20ms
```

```
Reply from 2404:6800:4000:1025::64: time=20ms
```

```
Ping statistics for 2404:6800:4000:1025::64:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
```

```
Approximate round trip times in milli-seconds:
```

```
    Minimum = 20ms, Maximum = 20ms, Average = 20ms
```

Task 4:- In your router's admin panel, locate the section where you can view or change the DHCP range. Write down the current range, and explain in 2-3 lines what would happen if two devices are assigned the same IP address in this range.

HCP Range: 192.168.29.X — Check the exact Start IP and End IP in your router settings.

If two devices are assigned the same IP address, an IP address conflict occurs. This can cause intermittent connectivity, loss of internet access, or difficulty communicating with other devices on the network.